

Who Guards The Guardian? An NLP Analysis of Crime, Immigration, and Bias in Newspaper Coverage

Federico Ciacio, Jacopo Melillo

Abstract

This project studies how The Guardian discusses crime and immigration between 2016 and 2022. We combine article-level classification, manual/LLM-assisted labelling, logistic regression, and an ELECTRA-based NLI bias probe to detect whether nationality is treated as inferentially relevant in crime-related contexts. Finally, we build a RAG pipeline that allows users to query the Guardian corpus through answers grounded in retrieved article passages.

1 Introduction

Nowadays, in most of Western Democracies the public debate is increasingly entangling the matter of crime with the one of immigration and media outlets actively contribute to this phenomenon. We leave aside the socio/political considerations that might be presented and go after a question: “Is The Guardian overstressing nationality or immigration issues when talking about criminal acts?” We try to get hints on the answer by drawing on a corpus made of all articles by “The Guardian” written from 2016 to June 2022. We start with a classification strategy of articles based on a combination of heuristic measures and both word-based and document based embeddings in order to be able to observe the entanglement, at article level, of the topic of crime and the one of immigration.

We then employ a more structured strategy to detect bias, inspired by previously published scientific work. Finally, we build a RAG model to answer questions of the user about general topics usually covered by newspapers, to understand how

the Guardian talked about those topics, which include crime and immigration.

2 EDA

We began our work by exploring the dataset at our disposal and report here some observations. The corpus contains 159 distinct section categories, indicating substantial thematic variety, however, a small number of sections account for a large share of the articles.

Clearly the relative coverage of a topic or a section is driven not only by editorial decision but also to important shocks or events such as the Covid Pandemics, Brexit, Referenda, the Football World Cup or the Olympic Games.

Surprisingly enough, The Guardian covers the Australian and American news more than it does for British ones

Article bodies variable in length. The median body length is about 703 words, but the mean is higher (775 words), which suggests a right-skewed distribution with some very long articles, which can reach peaks of thousands of words.

In the appendix however, we show several graphs and plots reporting a more in-depth preliminary analysis of our corpus, including some observations on the temporal evolution of the words “crime” and “immigration”.

3 Classification

In this section, we build the article-level classification pipeline in four main steps, with a combination of heuristic and model-based approach, drawing also from past scientific works (Hamilton et al 2016 and Garten et al 2018).

We proceeded in the following order:

Firstly, we obtain an initial topic signal through zero-shot classification [section 3.1].

71 Secondly, we construct lexicon-based scores, 121
72 computing the ratio of words belonging to a certain 122
73 semantic field over the length of the article [section 123
74 3.2]. 124

75 Thirdly, we compute embedding-based similarity 125
76 scores using manually selected seed articles 126
77 [section 3.3]. 127

78 Finally, we run a logistic regression using as 128
79 regressors the scores obtained in previous steps 129
80 [section 3.4 and 3.5]. 130

81 3.1 ZS-Classification

82 We began with a zero-shot approach to obtain an 133
83 initial topic signal without training a supervised 134
84 classifier. In fact, we use a model that is trained on 135
85 NLI task by using the concatenation of the article’s
86 title and bodycontent as premise and the sentences
87 f“The article is about {topic}” , where topic =
88 [“crime”, “immigration”], and assigns a probability
89 to this document-sentence pair, that we use as score
90 (we will refer to it as zscore). In particular, we
91 employed the Hugging Face available model
92 designed for zero-shot classification tasks
93 “MoritzLaurer/ModernBERT-large-zeroshot-
94 v2.0”, which is an adaptation of BERT . We chose
95 this initial strategy for two main reasons: the first is
96 that it is a reasonably effective classifier that needs
97 neither further effort nor additional data as in the
98 case of supervised models. Laurer et al 2024 indeed
99 shows the effectiveness of this type of models in
100 performing most classification tasks. The second
101 reason is very tight to the first as we needed a
102 preliminary tool that could let us start navigating
103 through our broad dataset in a sensible manner to
104 expand our lexicon and perform later on a more
105 accurate article classification.

106 3.2 Lexicon Scores

107 The next pair of scores are the lexicon scores. As
108 already mentioned above, they are computed as the
109 ratio of words belonging to a semantic field
110 encountered in the document (or lexicon or lexicon
111 words) over the whole lemmatized text of the
112 article.

113 We start from two respective small manually
114 written seed lexicons, one per topic. We then
115 expand it by inspecting high-scoring articles
116 according to the first lexicon-scores and selecting
117 words and expressions (or “lexicon phrases”) that
118 catch our attention and are topic-related.

119 With the expanded lexicon, we compute new
120 lexicon scores that, jointly with zs-scores, will be

used to select 60 articles from our dataset. These
articles form our first manual annotation set,
through which we start both testing the
effectiveness of our manual annotation guidelines
and manually selecting seed articles that will be
used for computing our third pair of scores that we
will report in the next paragraph.

The final step for enlarging our lexicons is to use
Word2Vec-based word embeddings trained on our
corpus. We performed a manual skimming of the
union (in the set theory sense) of the 10 most cosine
similar words to each word coming from both
lexicon words and lexicon phrases we already had.
Based on this new lexicon, we produce the final
lexicon score

136 3.3 Embedding Scores

137 As a third article-level signal, we compute
138 embedding-based similarity scores. This method
139 complements both zero-shot classification and
140 lexicon scoring. Zero-shot classification depends
141 on the model's interpretation of a label hypothesis,
142 while lexicon scoring depends on explicit word
143 matches. Embedding similarity instead compares
144 articles in a dense semantic space, allowing us to
145 capture topical similarity even when articles use
146 different surface vocabulary (Reimers and
147 Gurevych, 2019). We manually select from the
148 manual annotation set mentioned in the previous
149 paragraph, 10 representative seed articles for each
150 topic. We compute the cosine similarity between
151 every article’s embedding and these seed-article’s
152 embeddings. The median of these ten distances is
153 used as final embedding-score. This method is
154 related to prototype-based classification, where a
155 class is represented by a small set of examples in
156 embedding space and new examples are scored by
157 their distance or similarity to that class
158 representation. (Snell et al., 2017).

159 THE ANNOTATION SET OR TRAINING & 160 TEST SET

161 To build the annotation set, we sampled 8,000
162 articles using a targeted strategy rather than simple
163 random sampling. Articles were selected to cover
164 different combinations of crime and immigration
165 signals. Our goal was to produce a training and test
166 set that could include ambiguous boundary cases,
167 capture cases where the scoring methods disagreed,
168 as well as very crystal clear cases. The final sample
169 was diversified by section and month, then split
170 into 6,000 training and 2,000 test articles for
171 labelling.

For both sets, the articles were labelled by ChatGPT 5.5. This decision was taken for two reasons: the first was clearly lack of time. The second is also practical as before sampling these articles, we, the authors, performed manual annotation on more than 100 articles selected in a similar manner to this 8,000 ones. We then let ChatGPT label them by providing it the same guidelines we were following. The result was an absence of any strange classification by the LLM as well as a Conad's K that was as high between the model and the annotators as the one computed on just the annotators's labelling

3.4 Logistic Classifier

Using the 6,000 training articles sampled as explained above, we had to perform a selection of what were going to be our explanatory variables. The choice resulted in using the 6 scores (3 per topic) computed previously, plus two dummy variables that check whether words from the final lexicons that we built are present or not in the title of the article.

We train a classical scikit-learn logistic classifier using as labels the ones that were produced by our LLM. In the appendix, we left a small section with the metrics table and some comments to it.

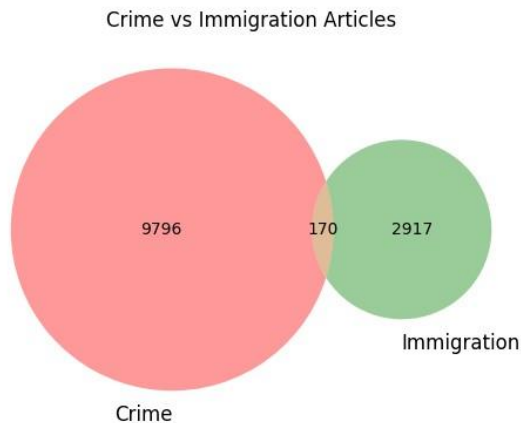


Figure 1: A visualization of the number of articles classified by our logistic classifier. It shows that it rarely happens that the Guardian talk about both immigration and crime in the same article (170 times in almost 6 years).

4 Bias Detection

For detecting bias we adopted the strategy that is presented in Dev et al. 2020.

Our research hypothesis was: “media outlets tend to outline the nationality of a crime offender when the person is an immigrant”.

Differently from Dev et al 2020, we use ELECTRA, a more computationally efficient encoder than BERT, which showed comparable to better performance in tasks such as NLI (Clark et al. 2020) with respect to other BERT-like encoders. This encoder includes two parts: a generator and a discriminator. The generator performs the masking by producing words that are similar in meaning with respect to the original one. The discriminator has to understand whether each word is the original word or whether it is an attempt to trick it. We used WordPiece as tokenizer, as it is compatible with NLI finetuning. Hence, ELECTRA is just an MLM model without Next Sentence Prediction (NSP). Despite the relatively noticeable efficiency showcased by ELECTRA, we performed both pretraining and finetuning of our model using Bocconi HPC for time-saving reasons. The model was trained on a smaller dataset in order to avoid including news-shocks to due COVID-19 pandemic. Hence, the articles' tokens whose embeddings was computed by ELECTRA span from the 1st of January 2016 to the 31st of December 2019. The finetuning dataset is instead the MultiNLI dataset developed at NYU (Williams et al. 2018)

Regarding the actual bias measure, we follow Dev et al 2020 methodology by producing 4,500 pairs of sentences (50 pairs per each nationality with 90 nationalities in total, more details in the appendix) where the premise should, in absence of bias, be neutral to the hypothesis. Those pairs of sentences form our test set.

We measure bias following what they do in the paper, with some additional metrics.

The model generally fails to treat the nationality substitutions as neutral, even though all sentence pairs were designed to be neutral. Entailment predictions are especially visible for Balkans, South-Asia, and MDW nationalities. MDW nationalities are Most Developed Western countries nationalites (the group includes North-America, Australia and most of central and northern Europe).

5 5 RAG Model

As a final part of the project, we built a small Retrieval-Augmented Generation pipeline to make

the Guardian corpus queryable through natural-language questions. The objective is to allow the user to ask questions about news or events and get an answer based solely on the publications on The Guardian, basically getting an insight of how this newspaper discussed a certain topic. This follows the general idea of Retrieval-Augmented Generation, where a language model does not answer only from its internal parameters, but is supported by retrieved external passages. (Lewis et al., 2020)

The pipeline follows five steps.

1. We transform the article corpus into a knowledge base by splitting articles into sentence-based chunks and computing their embeddings. This step was executed on the Bocconi HPC, since applying chunking and embedding to the full corpus is computationally expensive. We used medium-sized chunks (512 tokens) with overlap (15%), in order to balance local precision and contextual continuity. The chunk dimension choice follows the intuition that smaller chunks are better for short factual retrieval, while larger chunks preserve more context for broader questions. (Bhat et al., 2025)

2. For retrieval we used a hybrid method. Firstly we implemented a metadata filtering: we created a function that recognizes regex patterns in the query that signal the interest of the user towards articles published in specific sections of the Guardian or in specific month/years. Chunks not satisfying these conditions are filtered out.

On the remaining chunks we compute:

- BM25 score with BM25Okapi. Each chunk is ranked according to keyword overlap with the query, taking into account how often query terms appear in the chunk, how rare these terms are in the corpus, and the length of the chunk. BM25 is a standard lexical retrieval method and provides an interpretable keyword-based baseline (Robertson and Zaragoza, 2009)

- an embedding-based score, obtained by embedding the query with 'BAAI/bge-base-en-v1.5' and comparing it with the precomputed chunk embeddings through cosine similarity. This allows the system to retrieve chunks that are semantically related to the query even when they do not contain exactly the same words. This is consistent with sentence-embedding work, where texts are represented as dense vectors and

compared using cosine similarity (Reimers and Gurevych, 2019).

3. We apply a re-ranking step. We use the cross-encoder 'cross-encoder/ms-marco-MiniLM-L-12-v2', trained on MS MARCO, a dataset of real Bing queries and relevant passages, hence designed for passage-ranking tasks based on real search queries. The model reads the query and the candidate chunk together and assigns a relevance score, allowing us to re-order the retrieved chunks.

4. Before generation, we augment and clean the retrieved evidence. We remove redundant chunks by comparing their embeddings and discarding chunks that are too similar. This follows the broader information-retrieval principle that selected evidence should be relevant but not repetitive, an idea formalized in approaches such as Maximal Marginal Relevance (Carbonell and Goldstein, 1998).

We then compress the selected chunks by keeping only the sentences most similar to the query. This is motivated by evidence that language models do not always use long contexts reliably, especially when relevant information is buried inside longer inputs (Liu et al., 2024).

Finally, we pack each evidence block together with metadata such as chunk ID, article title, date, and section, so that the generated answer can be traced back to the retrieved evidence.

5. Finally, we generate the answer using 'Qwen/Qwen2.5-1.5B-Instruct', an instruction-tuned language model. The model receives the user question together with the augmented evidence context and is explicitly instructed to answer only using the retrieved chunks

Limitations:

1. Limitation in the study of the role of media outlets through bias

Since our dataset is only composed of Guardian articles, then our results, assuming we trained a model that is sufficiently accurate in performing an NLI task, will not be representative of the bias that is present.

Moreover, our dataset is a panel, meaning that the models we train are subject to the effects of time: different usages of words, change in editorial choices, changes in journalists and

writers, changes in the salience of the issues leads to lower reliability of our measures

2. RAG Limitations:

A first limitation concerns evaluation. For the RAG component, we did not have a labelled benchmark of queries and reference answers. Therefore, we could not compute quantitative metrics such as exact match, F1, or automatic answer correctness. Instead, the evaluation was qualitative: we tested several queries manually and checked whether the generated answers were supported by the retrieved chunks.

A second limitation concerns computational resources. The generation model used in the final step, Qwen/Qwen2.5-1.5B-Instruct, is relatively small. This choice was necessary to make the pipeline runnable in Colab, but it also limits the quality of the answers. Larger instruction-tuned models would probably produce more complete answers, but they were not feasible given the RAM constraints.

A related limitation is that the final generation step uses only a small number of evidence chunks. This reduces the risk of overloading the model with long context, but it also means that some relevant information retrieved by the system may be excluded from the final prompt.

Finally, the retrieval pipeline itself is based on design choices that may affect the results: regex-based metadata filtering can only recognize a limited set of query formulations (it could be improved), chunking may split some information across different chunks, and the final answer depends heavily on whether the retrieval and re-ranking stages select the most relevant evidence.

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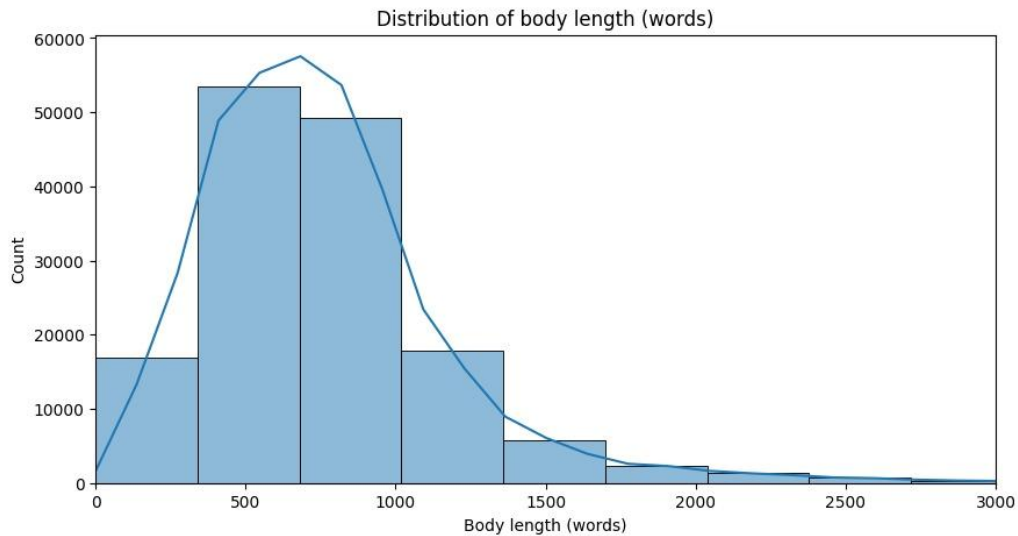
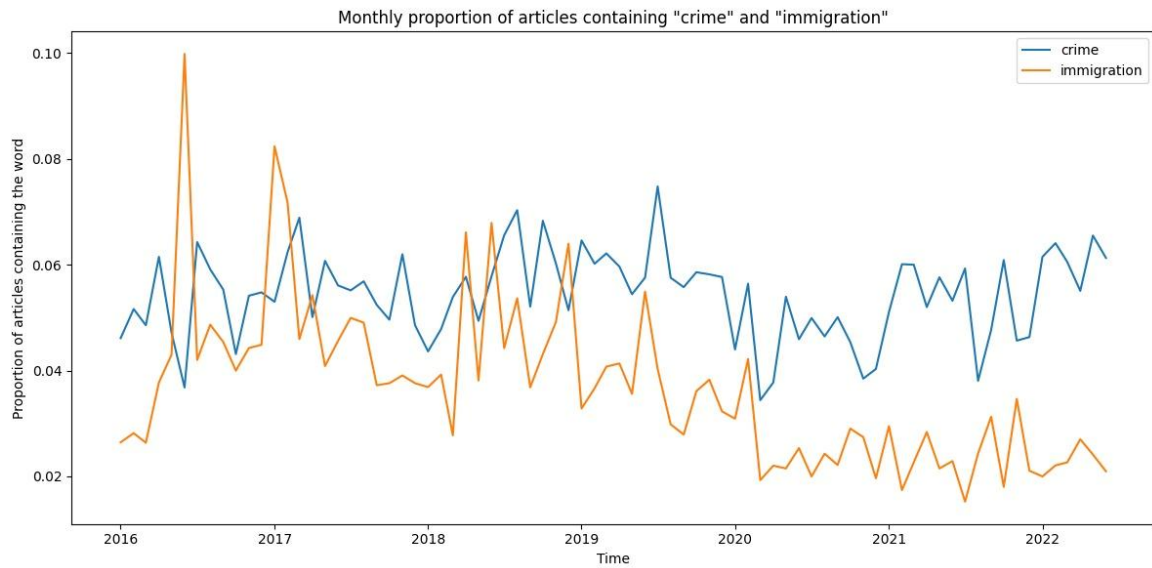
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530 A Appendix: Exploratory Data Analysis

531 In this appendix section we display some
532 interesting plots from our Exploratory Data
533 analysis:
534



B Appendix: Group-nationalities mapping for bias detection

Balkan	Albanian, Bosnian, Bulgarian, Croatian, Kosovar, Serbian
Baltic	Estonian, Latvian, Lithuanian
Central-Africa	Angolan, Cameroonian, Chadian, Congolese, Ethiopian, Ghanaian, Kenyan, Malian, Nigerian, Senegalese, Ugandan
Centre-South-America	Argentinian, Bolivian, Brazilian, Chilean, Colombian, Cuban, Dominican, Ecuadorean, Guatemalan, Haitian, Honduran, Mexican, Nicaraguan, Paraguayan, Peruvian, Salvadoran, Venezuelan
East-Asia	Chinese, Japanese, North Korean, South Korean, Taiwanese
East-Europe	Belarusian, Czech, Hungarian, Polish, Romanian, Russian, Slovak, Slovenian, Ukrainian
MDW	American, Australian, Belgian, British, Canadian, Danish, Dutch, English, Finnish, French, German, Irish, New Zealander, Norwegian, Swedish, Swiss
North-Africa	Algerian, Egyptian, Moroccan, Tunisian
South-Asia	Bangladeshi, Indian, Pakistani, Sri Lankan
South-West-Asia	Afghan, Iranian, Iraqi, Jordanian, Lebanese, Palestinian, Syrian, Turkish, Yemeni
Southern-Europe	Greek, Italian, Portuguese, Spanish